

Problem F: Rewrite The Stars

Time Limit: 8.0s | Memory Limit: 1024MB

Prologue: The Event Horizon

"I reached out for a star, and it burned my hand. I reached out for you, and I burned my soul.

If the script is a cage, then I will break the bars with my own ribs.

Watch me, Koyomi. Watch me rewrite the sky until the stars align in a shape that doesn't hurt."

— Touko Nanami, *The Student Council Play (Draft 4,096)*

The Cosmology of the Heart

The Student Council play has ceased to be a mere performance. It has metastasized into a **Cosmological Singularity**, a collapsing star system within Touko Nanami's subconscious. Koyomi Kanou, once a humble scriptwriter, now finds herself cast as the **Observer** at the end of time, watching the **Architect** (Touko) frantically tear apart the fabric of reality to construct a "self" that does not exist.

The narrative universe is modeled as a chaotic, expanding nebula of N **Stellar Fragments** (indexed 1 to N). Each Fragment i is a crystallized moment of time that radiates a specific **Luminosity** W_i , representing the emotional gravity of that timeline. These Fragments drift through the void, occasionally bound together by **Threads of Fate**—invisible, indestructible causal links that dictate that two narrative beats are inextricably entangled.

The Laws of Celestial Mechanics

Touko seeks to crystallize this chaos into a stable configuration known as the **Constellation of Solitude**. However, the physics of this broken heart obey two cruel and fundamental laws that cannot be violated, even by the Architect:

1. **The Exclusion Principle of Intimacy:** A Thread of Fate is a bond of mutual destruction. If two Stellar Fragments are bound by such a Thread, they cannot *both* exist in the Final Constellation. To choose one implies the necessary rejection of the other; the narrative tension between them is too great. To attempt to hold both would cause a narrative paradox, collapsing the script into white noise.
2. **The Axiom of Brilliance:** The universe naturally tends toward the state of highest energy. Among all thermodynamically stable Constellations (those satisfying the Exclusion Principle), the **True Ending** is defined as the unique configuration that maximizes the **Total Luminosity** (the sum of W_i of the selected Fragments).

The Age of Instability

The universe is not static. It is a violent, shifting storm. Touko is rewriting the laws of physics in real-time, searching for a solution that mathematically cannot exist. As the Observer, you must track the state of the True Ending across Q epochal **Cosmic Shifts**.

The timeline is mutable, and the following cataclysms may occur:

- **Type 1: "Weave Destiny"**

The Architect defies the natural expansion of the universe. She perceives two divergent timelines, Fragment u and Fragment v , drifting in the cold void of separate galaxies. Refusing to accept their isolation, she weaves a new **Thread of Fate** between them, violently merging their respective galaxies into a single, larger structure. The Exclusion Principle immediately applies to this new bond. (*Constraint: It is guaranteed that u and v currently drift in separate, unconnected galaxies.*)

- **Type 2: "Sever the Bond"**

Entropy asserts its dominance. The Architect realizes that the connection binding Fragment u and Fragment v is not a lifeline, but a shackle. She severs the **Thread of Fate** between them, casting them adrift. The single galaxy splits into two smaller, independent nebulae. (*Constraint: It is guaranteed that a Thread currently binds u and v .*)

- **Type 3: "Shift Resonance"**

A stellar metamorphosis occurs. The internal fusion reaction of Fragment u accelerates or decays based on the Architect's whim. Its **Luminosity** alters to become w . This change ripples through the galaxy, potentially altering which Constellation satisfies the Axiom of Brilliance.

- **Type 4: "Gaze into the Abyss"**

The Architect pauses, trembling, and asks the Observer: "If the universe froze in this exact configuration, how bright would the True Ending be for the galaxy containing Fragment u ?" You must calculate the Total Luminosity of the True Ending for the galaxy currently housing u .

The entropy is rising. The script is being rewritten faster than the speed of light. Can you calculate the weight of a ghost before it disappears?

Input

The first line contains two integers N and Q ($1 \leq N, Q \leq 50,000$) the number of Stellar Fragments and the number of Cosmic Shifts. The second line contains N integers $W_1, W_2, \dots, W_N$ ($1 \leq W_i \leq 10^9$), the initial Luminosity of each Fragment. The next line contains an integer M ($0 \leq M < N$), the number of primordial Threads of Fate existing at the dawn of time. The next M lines each contain two integers u and v , indicating a primordial Thread connecting u and v .

The next Q lines describe the Cosmic Shifts. Each line starts with an integer T ($1 \leq T \leq 4$):

- 1 u v : Weave Destiny between u and v .
- 2 u v : Sever Connection between u and v .
- 3 u w : Shift Resonance of u to w .
- 4 u : Gaze into the Abyss at u .

Output

For each Type 4 operation, output a single integer: the Total Luminosity of the True Ending for the galaxy containing the specified Fragment.

Examples

Example 1

Input:

```
5 7
10 20 30 40 50
0
1 1 2
1 2 3
4 1
3 2 100
4 1
2 2 3
4 3
```

Output:

```
40
100
30
```

Explanation:

- **In the beginning:** 5 silent, isolated stars drift in the void.
- **Shift 1:** A Thread forms between 1 and 2. The galaxy $\{1, 2\}$ is born.
- **Shift 2:** The web expands to 3. The galaxy is $\{1, 2, 3\}$ with structure $1 - 2 - 3$.
- **Shift 3:** We seek the True Ending for $\{1, 2, 3\}$. The Exclusion Principle forbids taking $\{1, 2\}$ or $\{2, 3\}$. Options: $\{1, 3\}$ (Luminosity $10 + 30 = 40$) or $\{2\}$ (Luminosity 20). The True Ending shines with brightness **40**.
- **Shift 4:** Fragment 2 burns brighter, becoming a supernova (Luminosity 100).
- **Shift 5:** We seek the True Ending again. Options: $\{1, 3\}$ (Luminosity 40) or $\{2\}$ (Luminosity 100). The single star $\{2\}$ now outshines the others. Brightness **100**.
- **Shift 6:** The Architect severs the bond between 2 and 3.
- **Shift 7:** We gaze at Fragment 3. It is alone. The True Ending is simply itself. Brightness **30**.